| | Document No. | Section No. |
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| | Effective Date: | Revision No. |
| | **PROCEDURAL MANUAL** | |
| | **MATERIAL HANDLING AND STORAGE** | |

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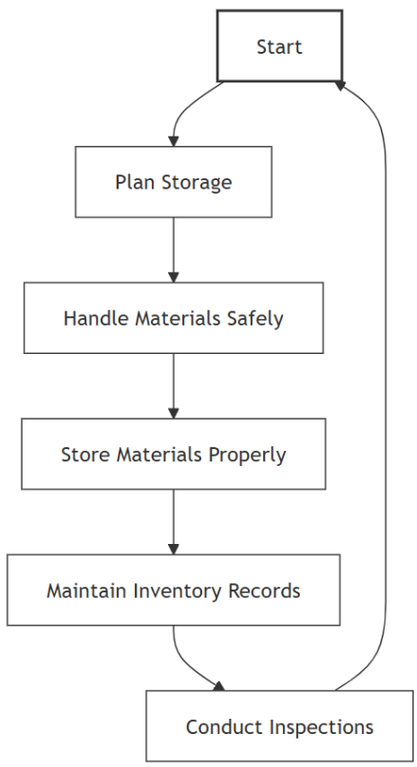
| **OBJECTIVE** | This procedure outlines the process for storing and handling construction materials on BHBI project sites and in warehouses to prevent damage, deterioration, loss, and safety hazards. |
| **SCOPE** | This procedure applies to all construction materials used on BHBI projects, including materials stored on site and in warehouses. |

**Responsibilities**

- **Project Engineer:** Responsible for overseeing material storage and handling on their projects.
- **Site Engineer:** Responsible for ensuring that materials are stored and handled properly on site.
- **Warehouseman:** Responsible for storing and handling materials in the warehouse (or designated on-site storage areas), under the supervision of the Site Engineer.
- **Foreman:** Responsible for assisting the Site Engineer in ensuring proper material storage on site.
- **All Site Personnel:** Responsible for handling materials carefully and reporting any issues related to material storage.

**Flowchart**




**Procedure**

**4.1. Material Storage Planning:**

- **Site Layout:** The Project Engineer, in consultation with the Site Engineer, designates specific areas on the project site for material storage, considering site access, laydown space, proximity to work areas, and safety.
- **Warehouse Layout:** If a warehouse is used, ensure it is organized to facilitate efficient storage and retrieval of materials, with designated areas for different material types.
- **Storage Requirements:** Consider the specific storage requirements of different materials, such as:
  - **Weather Protection:** Some materials need to be stored indoors or under cover to protect them from rain, sun, or other weather elements.
  - **Temperature and Humidity Control:** Certain materials may require specific temperature or humidity conditions.
  - **Ventilation:** Ensure adequate ventilation in storage areas to prevent the buildup of fumes or gases.
  - **Security:** Implement security measures to prevent theft or unauthorized access to stored materials.

**4.2. Material Handling:**

- **Lifting and Moving:** Use appropriate lifting equipment (e.g., forklifts, cranes) and techniques when moving heavy or bulky materials. Ensure equipment operators are properly trained and certified.
- **Manual Handling:** When handling materials manually, use proper lifting techniques to prevent injuries. Provide training on safe lifting practices.
- **Transportation:** Secure materials properly during transportation to prevent shifting or damage.

**4.3. Material Storage:**

- **Segregation:** Store different types of materials separately to prevent damage or contamination.
- **Stacking:** Stack materials safely and securely, following recommended stacking heights and methods. Use pallets or dunnage to keep materials off the ground and improve ventilation.
- **Labeling:** Clearly label all stored materials with their identification, quantity, and date of receipt.

- **Hazardous Materials:** Store hazardous materials (e.g., fuels, chemicals, paints) in designated areas, following all applicable safety regulations and guidelines. Ensure proper containment, labeling, and availability of Material Safety Data Sheets (MSDS).
    - **Deliverables:** MSDS Sheets
- **Flammable Materials:** Store flammable materials in approved fire-resistant cabinets or storage areas, away from sources of ignition.

**4.4. Material Protection:**

- **Weather Protection:** Protect materials from weather damage by storing them indoors, under cover, or using waterproof tarpaulins.
- **Moisture Protection:** Store materials off the ground on pallets or dunnage to prevent moisture damage.
- **Pest Control:** Implement measures to prevent pest infestations in storage areas.

**4.5. Inventory Management:**

- **Inventory Records:** Maintain accurate inventory records of all materials in storage, including quantities, locations, and dates of receipt and issuance (Refer to the *Inventory Control Procedure*).
    - **Deliverables:** Material Inventory Records

**4.6. Safety:**

- **PPE:** Personnel handling materials must wear appropriate personal protective equipment (PPE), such as gloves, safety glasses, and steel-toed boots.
- **Housekeeping:** Maintain clean and organized storage areas to prevent tripping hazards and facilitate safe access to materials.
- **Fire Prevention:** Implement fire prevention measures in storage areas, including no smoking policies and the provision of fire extinguishers.

**4.7. Inspections:**

- **Regular Inspections:** The Warehouseman, under the supervision of the Site Engineer, conducts regular inspections of storage areas to ensure that materials are stored properly, that storage areas are safe and well-maintained, and that inventory records are accurate.
    - **Deliverables:** Storage Area Inspection Reports

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